

CMPS6730

Natural Language Processing

AY 2026/2027, Fall Semester

Project Milestone Report

Are We Cooks or Cooked?

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1 PROBLEM OVERVIEW

When a cook is missing an ingredient, what should they use instead, and does the answer depend on the cuisine? Given a recipe, its cuisine, and an ingredient to replace, we rank possible substitutes using only recipe data. A good substitute rarely appears in the same recipe as the original, but appears with the same partner ingredients. For example, butter might be replaced with cream in French cooking but with ghee in Indian cooking.

2 DATA

We combine four public recipe datasets:

1. **RecipeNLG** (Bień et al., 2020): 2,231,142 cooking recipes (RecipeNLG_dataset.csv).
2. **RecipeBox** (Lee, 2017): about 125,000 recipes scraped from various food websites.
3. **Food.com Recipes and Reviews** (irkaal, n.d.): 522,517 recipes from 312 categories, with cooking times, servings, ingredients, nutrition, instructions and more.
4. **Food.com Recipes and Interactions** (Majumder et al., 2019): 180K+ recipes and 700K+ recipe reviews covering 18 years of user interactions and uploads on Food.com (formerly GeniusKitchen).

Data is downloaded from these sources and processed into one central dataset that keeps only the relevant fields: Source, Source_id, Title, Cuisine and ingredients_raw. The ingredient column contains both normalized ingredient names (some datasets normalize them for us) and unnormalized names with units and quantities. We clean this column by filtering out units, preparatory words such as “chopped” or “minced”, plurals and other irrelevant words such as “knives”.

Many recipes also appear more than once, because the source datasets sometimes share a primary source such as Food.com. We remove repeated recipes in three steps:

1. Match recipes across the two Food.com datasets by ID and remove the copy.
2. Remove identical recipes, defined as having the same normalized title and ingredients (lowercase, with spaces and special characters removed).
3. Remove recipes with too few ingredients (fewer than 2) or with rare ingredients (appearing fewer than 5 times across all 3,109,943 rows).

This reduced 3,109,943 raw rows to 2,721,199 recipes and 37,532 unique ingredients.

3 METHOD

For testing, we will hand-build a few hundred known substitution pairs from cooking substitution charts. We compare three methods:

1. **Association-rule or co-occurrence score.** Association rules mined with Apriori (Agrawal & Srikant, 1994) or FP-growth (Han et al., 2000) using the mlxtend library, ranked by confidence and lift. We expect this to fail, since substitutes rarely appear together, which makes it a useful negative result.
2. **Item2Vec similarity.** Cosine similarity between the Item2Vec vectors (Barkan & Koenigstein, 2016) of the missing ingredient and each candidate.
3. **Contextual score.** Similarity to the missing ingredient combined with how well the candidate fits the remaining ingredients in the recipe.

Implementation is still in progress, so no variants have been implemented yet. Possible variants include:

1. **Second-order co-occurrence.** Instead of ranking ingredients by how often they appear with the query, compare their whole co-occurrence profiles (cosine similarity of PPMI vectors).
2. **Rule-profile similarity.** Instead of returning the consequents of {query} → {b}, represent each ingredient by the confidence of its rules and compare those profiles. This applies the idea of the first variant to association rules.
3. **Order-free Word2Vec.** Shuffle each recipe and use a context window covering the whole recipe (Mikolov et al., 2013). This is common practice in Item2Vec- and food2vec-style work.

3.1 EVALUATION

We check whether the correct substitute appears near the top of each method's ranked list, using Hit@k and mean reciprocal rank. The popularity and co-occurrence methods are the baselines. Outputs: a results table by method, a plot of accuracy by cuisine, and example substitutions (global vs. cuisine-specific).

4 PRELIMINARY EXPERIMENTS AND RESULTS

Data collection is complete. After cleaning and de-duplication, the dataset contains 2,721,199 recipes and 37,532 ingredients, down from 3,109,943 raw rows. We randomly split the recipes into training (70%), validation (15%) and test (15%) sets.

5 RELATED WORK

Agrawal and Srikant (1994) introduce Apriori, the standard algorithm for finding frequent itemsets and association rules, and define support, confidence and lift. It finds items that often appear together. It is our classical baseline, and we test whether its rules can also reveal substitutes.

Han et al. (2000) present FP-growth, which avoids Apriori's candidate generation. It matters for our project because it is the practical variant that scales at low support thresholds.

Mikolov et al. (2013) introduce Word2Vec, which learns word vectors from the words around them. We apply the same idea to recipes, treating ingredients as words.

Barkan and Koenigstein (2016) apply skip-gram to item sets (baskets) instead of sentences. It is the template for our embedding approach and differs from the rule-mining papers by learning dense item similarity rather than explicit rules.

Ahn et al. (2011) analyze ingredient pairing through shared flavor compounds and recipe co-occurrence across cuisines, and find that pairing rules differ between Western and East Asian cuisines. It is a chemistry-and-cuisine perspective on complements, in contrast to our purely statistical, substitution-focused methods.

Teng et al. (2012) build separate complement and substitution networks from Allrecipes recipes and reviews, and show that substitutes can be inferred from user-modified recipes. This is the closest prior work to ours. We learn substitutes from ingredient lists alone and compare cuisines.

Shirai et al. (2021) build a food knowledge graph and rank substitution candidates, evaluated against a curated set of known substitutions. It is a possible source of substitution ground truth for our test set.

Fatemi et al. (2023) use a graph neural network that ranks substitutes using the whole recipe as context. We use simpler methods and study cuisine as the context.

6 DIVISION OF LABOR

Member	Responsibilities
Tony	Data collection, aggregation and running algorithms
Martin	Running algorithms, report drafting and secondary research

7 TIMELINE

Date	Milestone
Thursday, 12 November 2026	All relevant data collated, secondary research complete, first milestone submitted
Friday, 20 November 2026	Base implementations of all methods complete and evaluations recorded
Friday, 4 December 2026	Improved implementations (variants) complete and evaluations recorded
Tuesday, 8 December 2026	Final submission

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